UX Research Questionnaire & Unsupervised Clustering System Design

This document defines the architecture for classifying insurance agents by behavioural pattern rather than by fixed demographic rules. Instead of hardcoding a target set of personas into the system, the architecture evaluates the full empirical response distribution and lets a downstream clustering algorithm discover whatever groupings actually exist in the data — in count, shape, and character.

1. Operational Vector Space Framework

Every respondent is mapped into a four-dimensional coordinate vector: R = [SUP, WP, AR, PR]. Qualitative journey-map answers and quantitative scale answers are both translated into this shared numeric space so that agents can be compared and grouped purely on mathematical distance and density, not on which demographic bucket they were pre-sorted into.

Dimension	Code	What it captures
System Utility Perception	SUP	Perceived baseline value, speed, and reliability of official digital tools and portals.
Workaround Propensity	WP	How often an agent substitutes personal tools (LINE, paper, spreadsheets) for official systems, and why.
Abandonment Rate	AR	How often an agent hits friction severe enough to abandon a task inside the official system — surfaced through pain-point text across the journey map.
Peer Reliance	PR	How dependent an agent is on senior colleagues or administrative staff to complete digital tasks, rather than doing them independently.

Demographic and tenure fields (Age, Position, Years of Experience) are deliberately excluded from R. They are retained in the dataset only as descriptive variables to be examined after clustering — e.g., “does this discovered cluster skew toward newer agents?” — never as inputs that define or constrain the clusters themselves.

2. Vector Formulas & Questionnaire Data Sources

Each dimension resolves to a specific, unambiguous set of questionnaire fields. No formula mixes unrelated question types, and no referenced field is descriptive-only data smuggled in as a behavioural signal.

System Utility Perception (SUP)

Formula: Mean score of Q8 (tool/system familiarity), Q10 (perceived efficiency impact), and all sub-items in Q11 (System & Digital Tool Impact matrix: Digital Tool Usage, System/Portal Speed & Reliability, Mobile App Experience).

Q11 is scoped to system/tool-specific items only. Professional-capability items (Sales Pitch, CRM, Customer Service, Insurance Knowledge, tracked separately in Q12) are excluded, since they measure skill rather than tool perception.

Workaround Propensity (WP)
Formula: Aggregated indexing of multi-select counts in Q13 (reasons for bypassing the company system), Q14 (personal-app/paper-calendar selections and stated reasons), and Q23 (Personal Tools / Workarounds list). Q23 lists only genuine personal tools/ workarounds. Official systems (Tou-Bao-Tong, Fu-Wu-Tong, Li-Pei-Tong, AP System, ABC Academy, Proposal System) live in a separate question (Q22) and are excluded from this formula.

Abandonment Rate (AR)
Formula: Normalised count of system dropouts and pain-point mentions, extracted via keyword/semantic tracking from Q27(c) — the pain-point field — across every action in all six journey-map tasks. Every action in Q27 uses the identical a) location / b) tools-people / c) pain-point structure, so this formula applies uniformly across Tasks I–VI without task-specific exceptions.

Peer Reliance (PR)
Formula: Frequency index of "Senior colleague" or "Administrative Assistant / Peer support" selections made under Q27(b) across all task actions, cross-validated against the equivalent option in Q23. "Administrative Assistant / Peer support" is available as an explicit option inside every Q27(b) tools/people list, so this signal is genuinely present everywhere the formula looks for it.

3. **Downstream Unsupervised Cluster Discovery Logic**
The analytics engine processes respondent vectors without structural preconceptions, predefined centroids, or a fixed target cluster count. The discovery sequence:
(1) Dimensionality Scaling & Normalisation — raw R = [SUP, WP, AR, PR] vectors are normalised so no single dimension dominates distance calculations due to scale differences.
(2) Density & Distance Distribution Profiling — the model evaluates global point density across the vector space to find natural groupings, rather than measuring distance from any hardcoded coordinate.
(3) Dynamic Cluster Identification — an unsupervised method (e.g., K-means with Silhouette-score optimisation, or density-based clustering such as DBSCAN) automatically determines how many clusters best fit the current dataset. This count is not fixed in advance and is expected to vary between survey waves as agent behaviour shifts.
(4) Emergent Behavioural Persona Tagging — qualitative labels are applied strictly after the fact. An analyst reviews each discovered cluster's centroid and its distinguishing pain points, tools, and journey patterns, then assigns a descriptive label to that specific cluster.

Operational note
Because cluster count and shape are data-driven, the number of personas the system

surfaces in any given analysis run may be more or less than five, and may not resemble prior research archetypes at all. The system must be able to represent, label, and visualise an arbitrary number of clusters — it cannot assume five slots to fill.

4. **Implications for Dashboard Implementation**
   A dashboard built on this design must treat cluster count, centroid values, and cluster labels as run-time outputs of the clustering job — never as compile-time constants. Concretely, the dashboard needs to support:
   - A variable number of persona/cluster cards — not a fixed set of five — rendered from whatever the current clustering run returns.
   - Centroid values (SUP/WP/AR/PR, and any displayed skill or work-allocation averages) are computed live from the respondents currently assigned to each cluster, not stored as fixed reference numbers.
   - Cluster labels and descriptive narratives (including any “why was this agent classified this way” explanation shown per agent) generated from that cluster's actual distinguishing features — its dominant pain points, tool usage, and vector position — rather than a pre-written narrative tied to a demographic band.
   - Demographic fields (age, tenure, position) surfaced only as descriptive cross-tabs within a discovered cluster (e.g., “68% of this cluster has 10+ years of experience”), never as the rule that assigned the agent to that cluster in the first place.
   - An editable labelling step for analysts, since final cluster names are a human judgment call applied after the algorithm runs — the dashboard should not assume the label is permanent or globally correct across every clustering run.